

Mushrooms, tumors, and immunity: an update.



There is significant interest in the use of mushrooms and/or mushroom extracts as dietary supplements based on theories that they enhance immune function and promote health. To some extent, select mushrooms have been shown to have stimulatory action on immune responsiveness, particularly when studied in vitro. However, despite their widespread use for potential health benefits, there is a surprising paucity of epidemiologic and experimental studies that address the biologic activities of mushrooms after oral administration to animals or humans. There have been a number of studies that have addressed the ability of mushrooms to modulate mononuclear cell activation and the phenotypic expression of cytokines and their cognate receptors. There have also been a number of attempts to determine antitumor activities of mushrooms. Such studies are important because many of the components of mushrooms do potentially have significant biologic activity. All data, however, should be tempered by the possibility that there are toxic levels of metals, including arsenic, lead, cadmium, and mercury as well as the presence of radioactive contamination with ^{137}Cs . In this review, we will present the comparative biology with respect to both immunological and antitumor activities of mushroom extracts and also highlight the need for further evidence-based research.